THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____
Seat Number _____
Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

School of Mathematics & Physics

Semester 2 Examinations, 2025

PHYS4055 Atomic Physics and Quantum Optics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

•Any written or printed material is permitted; material may also be annotated.
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions

Materials Permitted in the Exam Venue:
(No electronic aids are permitted e.g. laptops, phones)

Any additional written or printed material is permitted; material may also be annotated.

Materials to be supplied to Students:
Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

1 x 14-Page Answer Booklet

Instructions to Students:
If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

There are FOUR questions. Answer any THREE.

For Examiner Use Only

Question	Mark

Total _____

Question 1

(10 marks)

In this question, we look at the hyperfine splitting of levels, which occurs due to the interaction of atomic electrons with the nucleus.

We will consider the atom with electronic term $^2\mathrm{P}_{1/2}$ and nuclear spin $I = 5/2$.

(a) (1 mark) What are the possible values of the total atomic angular momentum F, where $\mathbf{F} = \mathbf{I} + \mathbf{J}$, with $\mathbf{I}$ and $\mathbf{J}$ the nuclear spin and electronic angular momentum?

(b) (2 marks) The Hamiltonian for the magnetic hyperfine interaction may be expressed as the scalar product of the nuclear spin and electronic angular momentum, $H = A\mathbf{I} \cdot \mathbf{J}$, where A is the magnetic hyperfine constant.

Show that the hyperfine energy shift is given by

$$\Delta E = \frac{A}{2}\big(F(F+1) - I(I+1) - J(J+1)\big) \,, \tag{1}$$

showing all working.

(c) (2 marks) Hence, find an expression for the energy *difference* between hyperfine levels for the considered term, and sketch an energy-level diagram.

(d) (3 marks) Consider that a *weak* magnetic field is applied to the atom.

(i) (1 mark) Show how your energy-level diagram from (c) is modified in this field, justifying your answer.

(ii) (1 mark) How many levels of different energy are there? Explain your result.

(iii) (1 mark) Describe, with the aid of a diagram, the pattern of the energy-level splitting if the magnetic field were *strong* rather than weak.

(e) (2 marks) As well as the magnetic hyperfine interaction, in general there is also the possibility of contributions to the hyperfine splitting from interaction of electrons with higher-order nuclear moments.

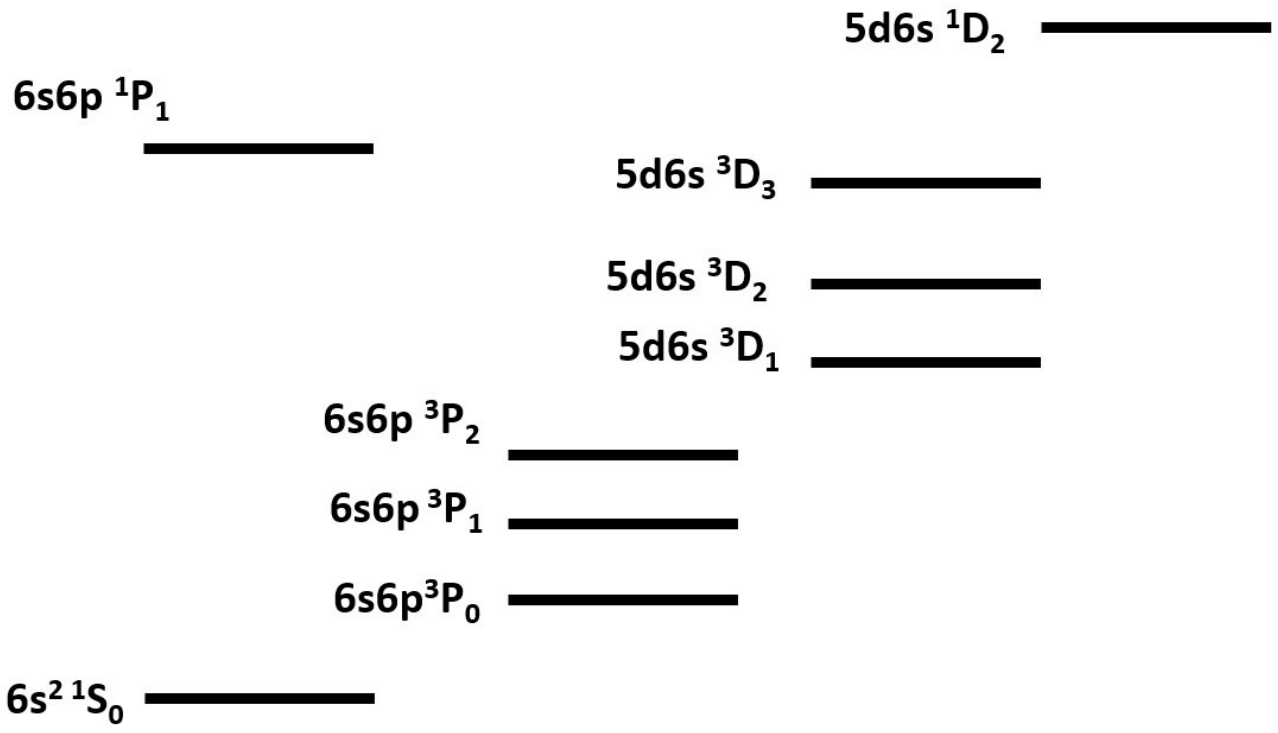
Using the Wigner-Eckart theorem, briefly explain why there will or will not be a contribution to the hyperfine splitting from higher-order nuclear moments for the atom considered in this question.

Next question over ...

Question 2

(10 marks)

(a) (2 marks) The figure below shows a partial level diagram of Yb ([Xe]$4f^{14}6s^2$). Identify all the E1-allowed transitions in this system and mark them on the diagram below.



(b) (1 mark) A transition between one of the excited states and the ground state can be used as a clock transition. Propose which one (there is more than one suitable transition, but one is sufficient). Explain your answer.

(c) (1 mark) In order to construct an atomic clock the atoms must be decelerated and trapped. Propose a suitable transition for this. Explain your answer.

(d) (4 marks) An atomic beam of Yb (M=1.73×10^{-25} kg) is produced in a source at 300 K and slowed by counter-propagating laser radiation, using the transition that you deemed suitable for laser-cooling. Assume that the wavelength of this transition is 399 nm and the lifetime of the excited level is 5.5 ns.

(i) (1 mark) What is the role of the magnet in the Zeeman slower?

(ii) (2 marks) What is the stopping distance at half the maximum deceleration?

(iii) (1 mark) How many photons are scattered to stop the atom?

$h = 6.62607 \times 10^{-34}\,\mathrm{J\,s}$; $k_B = 1.38065 \times 10^{-23}\,\mathrm{J/K}$

(e) (2 marks) After the Zeeman slowing, the atoms are cooled in optical molasses.

(i) (1 mark) Why are the lasers in optical molasses red detuned?

(ii) (1 mark) What is the Doppler-limit temperature for this atom, assuming the cooling is done on the transition discussed above?

Question 3

(10 marks)

A beam-splitter is an optical device with two input modes and two output modes. Consider a 50:50 beam splitter with input modes 1 and 2 and output modes 3 and 4. This beam-splitter combines the modes as $\hat{c} = \left(\hat{a} + \hat{b}\right)/\sqrt{2}$ and $\hat{d} = \left(\hat{a} - \hat{b}\right)/\sqrt{2}$.

(a) (3 marks) What output state is expected for the input state $|\Psi_{\text{in}}\rangle = \frac{1}{\sqrt{2}}\left(|2,0\rangle + |0,2\rangle\right)$?

(b) (2 marks) Calculate the probabilities P_{20}, P_{02}, and P_{11} of detecting 2 photons in $\hat{c}$, 2 photons in $\hat{d}$, or one photon in each output port.

(c) (2 marks) Briefly remark on the physical significance of the detection probabilities in the previous question.

(d) (3 marks) Suppose we instead input the state $|\Phi_{\text{in}}\rangle = \frac{1}{\sqrt{2}}\left(|1,1\rangle + |2,0\rangle\right)$. Derive the probability P_{11} of detecting one photon in each output mode after the beam splitter.

Question 4

(10 marks)

In a frame rotating at the optical frequency, a single optical cavity mode interacting with an external field through a cavity mirror can be described by the quantum Langevin equation: $\dot{\hat{a}} = -\gamma/2\ \hat{a} + \sqrt{\gamma}\hat{A}_{in}$, where $\hat{a}$ is the annihilation operator for the cavity mode, γ represents the coupling rate through the mirror and $\hat{A}_{in}$ is the annihilation operator representing the input field.

Now suppose we introduce a pumped non-linear crystal inside the cavity that induces the interaction Hamiltonian $H_i = i\hbar\frac{\chi}{2}(\hat{a}^2 - \hat{a}^{\dagger 2})$, where χ is a real number.

(a) (2 marks) Show that the quantum Langevin equation describing this cavity mode is now:

$$\dot{\hat{a}} = -\chi\ \hat{a}^\dagger - \gamma/2\ \hat{a} + \sqrt{\gamma}\hat{A}_{in}\ . \tag{2}$$

Show all working.

(b) (3 marks) Find the solution to the above Langevin equation for the amplitude quadrature of the cavity mode in the Fourier domain, $\tilde{X}^+ = \tilde{a}(\omega) + \tilde{a}^\dagger(\omega)$.

(c) (2 marks) Using the input/output relations, show that the amplitude quadrature of the output field is given by:

$$\tilde{X}^+_{out} = \frac{\gamma/2 - i\omega - \chi}{\gamma/2 + i\omega + \chi}\tilde{X}^+_{in}\ . \tag{3}$$

(d) (1 marks) Under what conditions will the spectral variance of the output field, $V^+ = |\langle\tilde{X}^+_{out}\rangle|^2$, be squeezed?

Suppose Alice and Bob, via some protocol, are able to share maximally entangled photonic qubits (i.e. Bell pairs) between them.

(e) (2 marks) By sequentially measuring the entangled photon pairs in the same basis they can generate a correlated string of bits. Describe why the resulting bit string is random and secret.

END OF EXAMINATION